UNIT 3

PUBLIC POLICY FOR FAIRNESS AND EFFICIENCY

A. Introduction

Policies, causality, natural experiments

Example: Women's vote and child health programs

Possible **causal** chain:

Naturally assigned treatment and control groups in the US:

- **Treatment**: States where women had been given the vote
- **Control**: Same states, just before women got the vote

→ Outcome: Increase in social services and spending on children, decrease in child deaths

Context

- **Institutions** play a key role in determining economic outcomes, including inequality and growth. (Unit 1)
- **Game theory** can help us model **social interactions** and predict economic outcomes. (Unit 2)
- **Preferences** and **policies** to address **social dilemmas** (Unit 2)

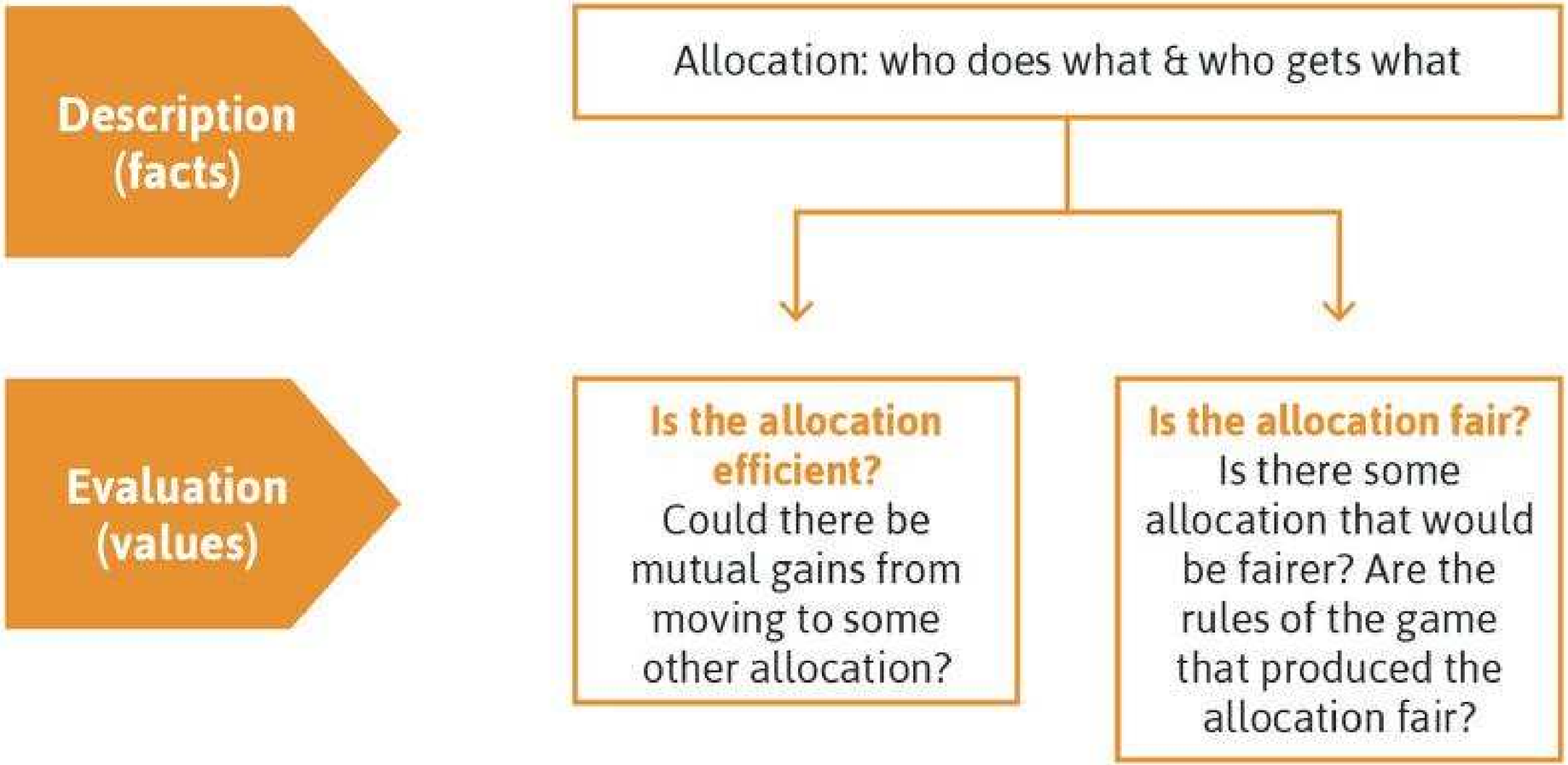
- Which economics outcomes are desirable?
- How can policies help us achieve these outcomes?

This Unit

- The concept of **efficiency** and a Pareto allocation
- Notions of **fairness**, and trade-offs that result
- **Policies** with unintended consequences

B. Evaluating Allocations

Evaluating economic outcomes



How can public policies improve observed outcomes?

Allocations and values

- **Efficiency**: There is no alternative technically feasible allocation in which at least one person would be better off, and nobody worse off.
- **Fairness**: Evaluation based on one's conception of justice

Many other evaluation methods we might use, such as:

- Individual freedom
- **Social welfare**, or maximizing *total* utility
- Social cohesion
- Conformity to religious beliefs

Social preferences in the ultimatum game

A Responder who cares only about own payoffs would accept any positive offer, because something is better than nothing.

A Responder who thinks that the Proposer's offer has violated a **social norm** of fairness, or that the offer is insultingly low for some other reason, might be willing to <u>sacrifice the payoff to punish</u> the Proposer.

Evaluating outcomes:

- If Responder rejects, the outcome—throwing away money!—cannot be efficient.
- **Dictator game** where Responder cannot reject offer eliminates this inefficiency, but is hardly fair!

→ There may be trade-offs between efficiency and fairness

C. Efficiency: The Pareto Criterion

“Efficiency” in economics

- An allocation is **Pareto efficient** if nobody can be better off without making somebody worse off.
- **Pareto criterion** can help us judge between two allocations
- A very unequal distribution of food—even one that means some people will starve—can be Pareto efficient as long as all the food is eaten by a person who enjoys it even a little.

→ <u>The Pareto criterion says nothing about fairness</u>

Recall: The pesticide game

When Anil and Bala each play their dominant strategy, the outcome is (Terminator, Terminator).

Anil and Bala each receive payoffs of 2, but both would be better off if they both used IPC instead.

<u>The predicted outcome is therefore not the best feasible outcome.</u>

		Bala	
		IPC	Terminator
Anil	IPC	3 / 3	1 / 4
	Terminator	4 / 1	2 / 2

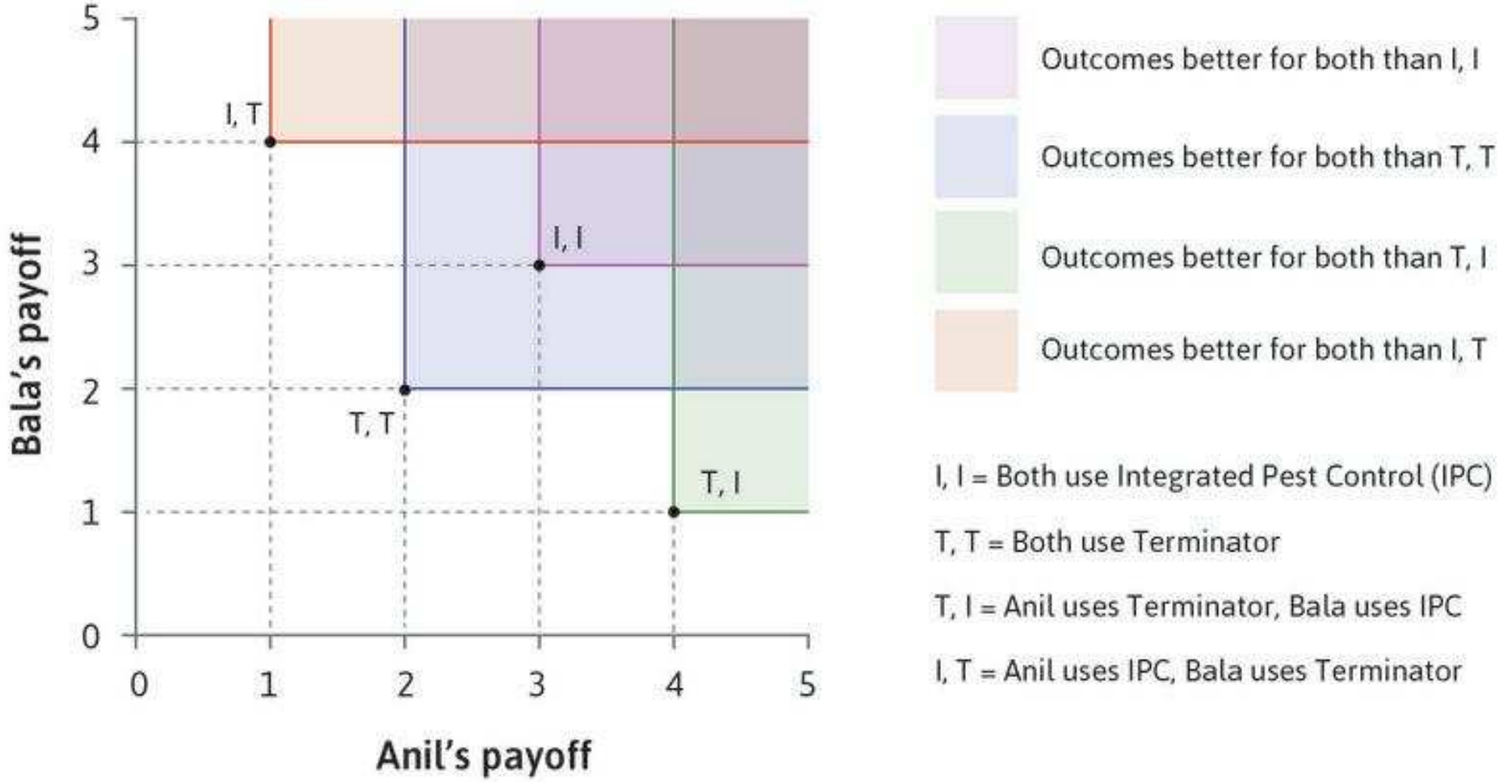
Efficiency in the pesticide game

(I,I) **Pareto-dominates** (T,T)

(I,I), (I,T), and (T,I) are all Pareto efficient.

Pareto criterion does not tell us which efficient allocation is better:

We cannot rank them.

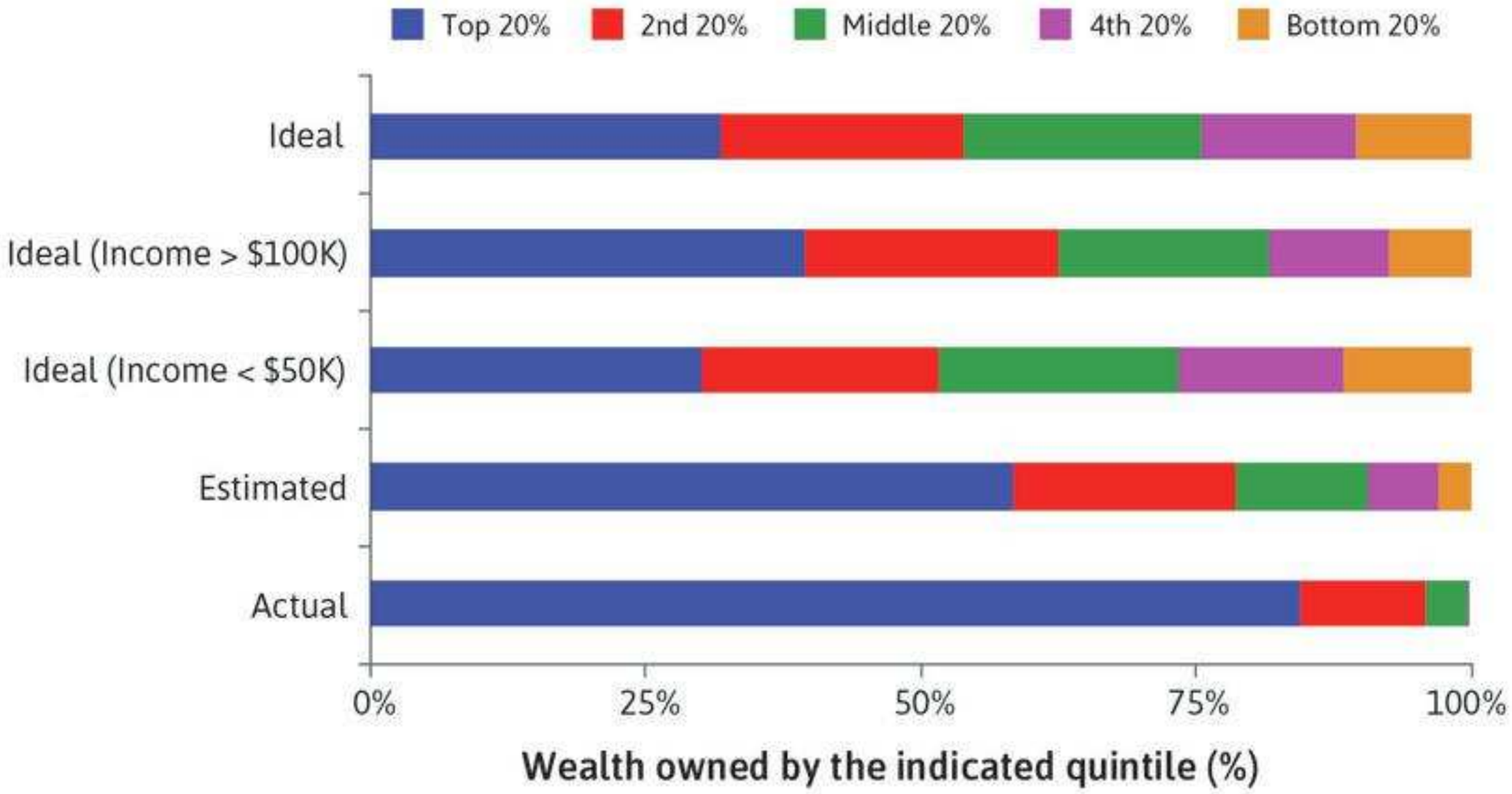


Anil playing IPC and Bala free riding by playing Terminator is Pareto efficient, but we (and Anil) may think this is unfair.

D. Fairness and Inequality

Too much inequality?

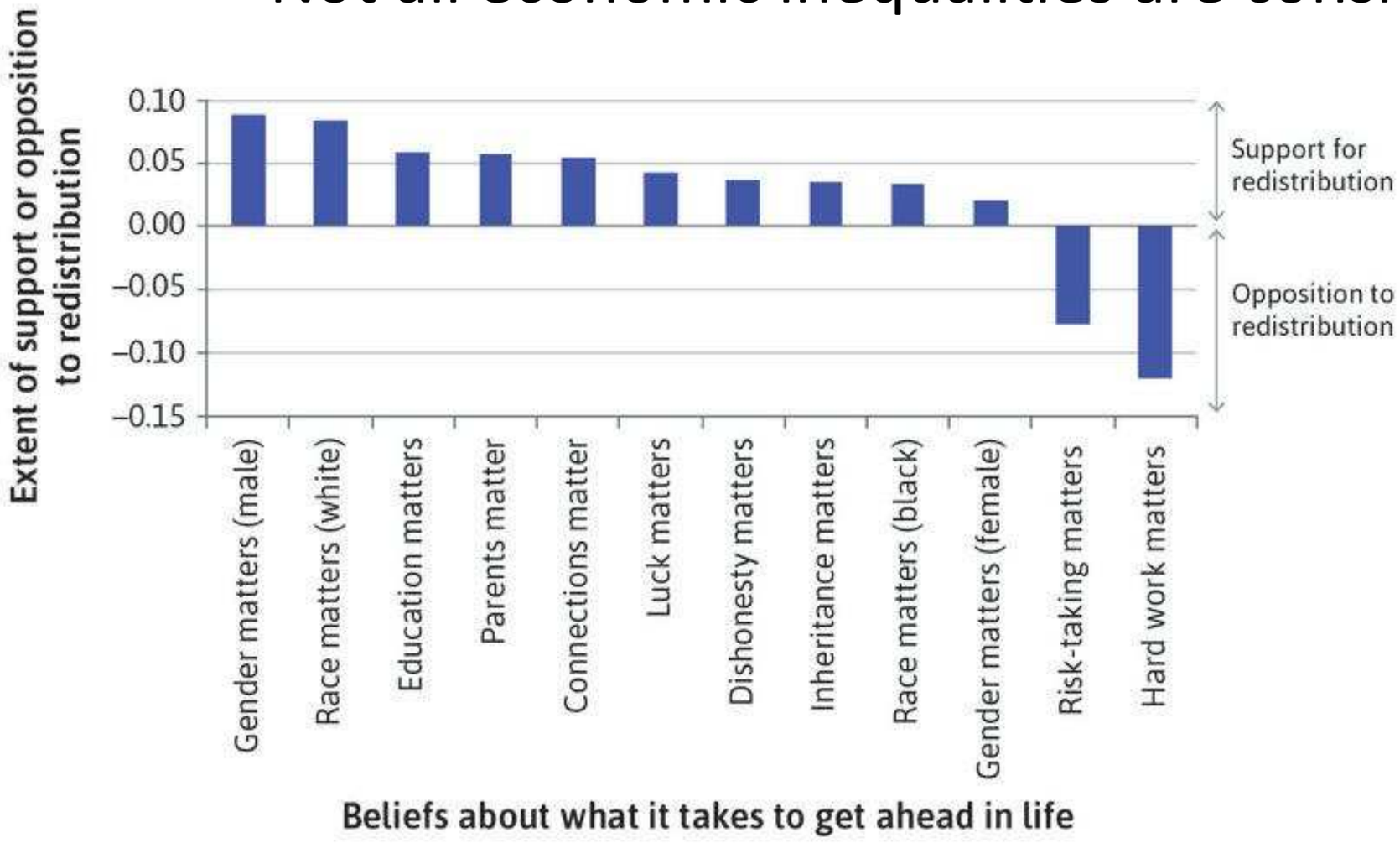
Americans' ideal, estimated, and actual distribution of wealth:



The actual distribution is much more unequal than the public's estimate – and contrasts sharply with the even lower inequality they would find ideal.

Fair inequality or a tilted playing field?

Not all economic inequalities are considered unfair.



A person who thinks that hard work and risk-taking are essential to economic success is much less likely to support pro-poor redistribution.

Defining fairness

Allocations can be judged unfair because of:

- *How unequal they are:* In terms of income, for example, or subjective well-being or the distribution of wealth. These are **substantive judgements of fairness**, which are based on the characteristics of the allocation itself.
- *How the inequalities came about:* For example, by force or racial discrimination, by competition on a level playing field, or by hard work. These are **procedural judgements of fairness**.

Procedural judgements of fairness

Evaluate rules of the game according to aspects such as:

- *Voluntary exchange through legitimate means*: Were the actions resulting in the allocation the result of free, lawful choices by the individuals involved?
- *Equal opportunity for economic advantage*: Or were they subjected to some kind of discrimination because of their race, sexual preference, gender, or who their parents were?
- *Needs*: Did the rules take account of the extent to which individuals need, or deserve, the amounts they get?

Questions of value in economics

No science can eliminate disagreements about questions of value. But economics can clarify:

- <u>How the dimensions of unfairness may be connected</u>: For example, how the rules of the game that give special advantages to one or another group may affect the degree of inequality.
- <u>The trade-offs between the dimensions of fairness</u>: For example, are some kinds of unfairness essential for achieving efficient outcomes?
- <u>Public policies to address concerns about unfairness</u>: Also, identifying and quantifying the (causal) effects of policies.

E. Public Policies, Unintended Consequences

Implementing public policies

- Outcome of a public policy is the result of an interaction between the government's actions and the privately chosen actions of those affected

- Policies can change economic outcomes by influencing what actions people decide to take. This is done by changing

  - Incentives: A policy changes the benefits or costs of alternative courses of action open to the individual.

  - The information available: People can use all available information when they make decisions about which actions to take.

Tragedy of the Commons

- A **prisoners' dilemma** in which overgrazing is the dominant strategy
- Overgrazing is the **Nash equilibrium** but restricting would support higher payoffs for both.
- An effective government policy might alter this situation by <u>changing the Nash equilibrium.</u>

		Farmer B	
		Restrict	Do not restrict
Farmer A	Restrict	10 / 10	6 / 12
	Do not restrict	12 / 6	8 / 8

A tax on overgrazing the commons

Impose a tax on cows, beyond a certain number of cattle.

→ Reduces the payoffs for 'Do not restrict'

→ (Restrict, Restrict) is new Nash equilibrium

		Farmer B	
		Restrict	Do not restrict
Farmer A	Restrict	10 / 10	6 / 8
	Do not restrict	8 / 6	4 / 4

Features of the overgrazing tax:

- It treats both herders equally.
- Each herder must take account of the cost that their overgrazing imposes on the other.
- Better herders find it more valuable to put their cows on the pasture: Those less skilled will not find it worth paying the tax.

… but Policies can affect Preferences

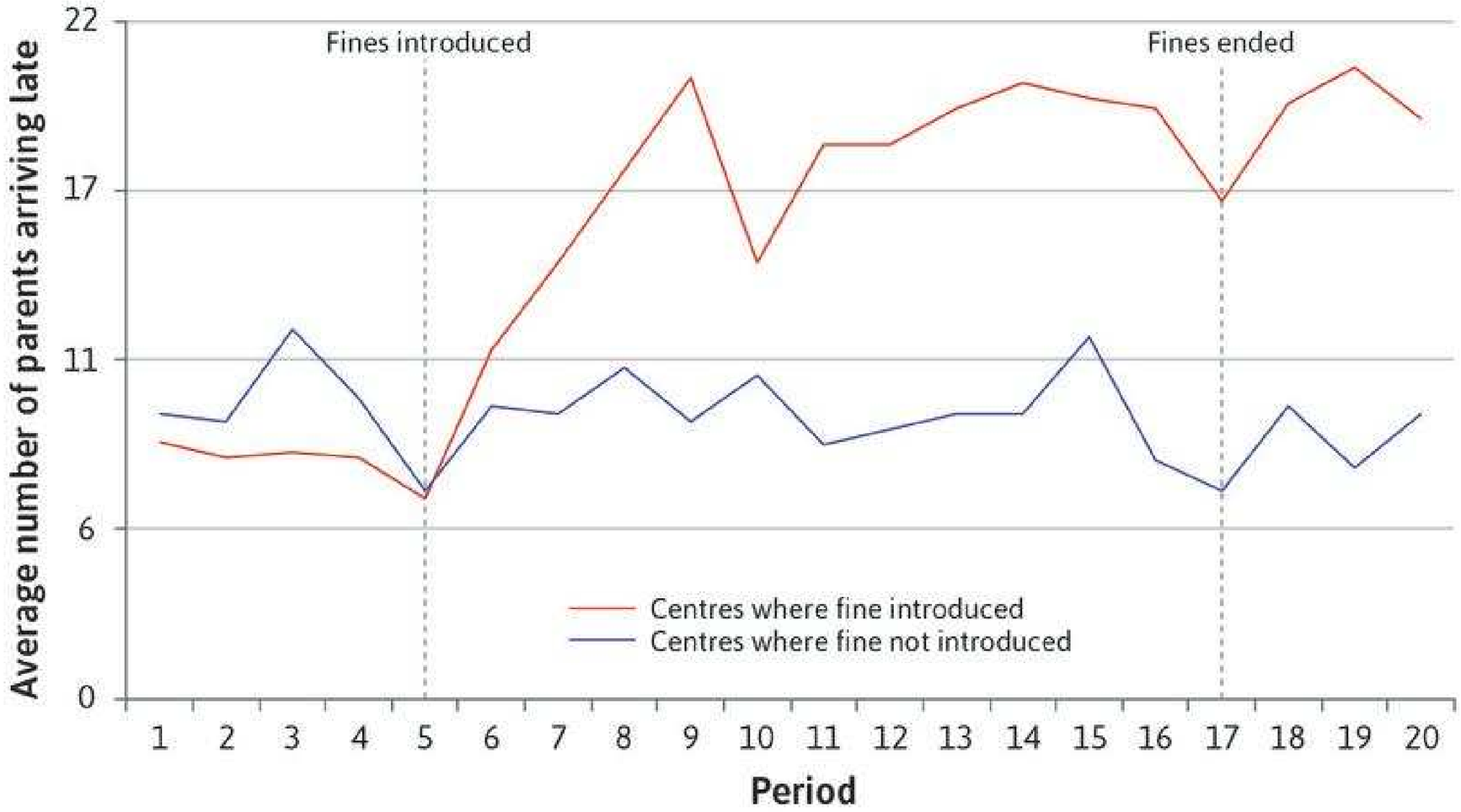
Example: Daycare centers in Israel

There was an issue with parents picking up their children very late.
This meant more staff had to stay on-site later in the day.

To deter this behavior, it was suggested to charge a late fee.

The late fee was tested at some daycare centers, but not all.
(Treatment and Control groups)

Treatment Failed – Response Persisted



The use of a market-based incentive (price of lateness) provided a new frame for the decision.

Economic incentives have **crowded out** social preferences.

What have these two examples shown?

In order to avoid unintended consequences, a policymaker must make sure that:

- The policy does not change people's preferences in unintended ways: A change in preferences might mean that behavior changes too, which may mean the policy fails to hit its target.
- The intended outcome is a Nash equilibrium: Under the new policy, people must be motivated to act in ways that are consistent with the objective of the policymaker.

Assessing the effect of Taxes

- Consumption data helps reveal how the amount sold varies in response to a change in price. A useful measure is the **price elasticity of demand**

  = the percentage change in demand resulting from a 1% increase in price, expressed as a positive number.

  Demand is *elastic* if this is greater than 1, and *inelastic* if less than 1.

- <u>Taxes on inelastic demand mostly raise government revenues</u>: For example, a tax on cigarettes. Smoking demand is highly inelastic, so substantial revenues could be collected.

- <u>Taxes on elastic demand mostly change consumer behavior</u>: Example, a tax on sweet foods. Sweets demand is very elastic, so this would mostly reduce consumption of sweets.

Policy evaluation: Further examples

Advertising strategy	Intellectual property rights
Aim: Increase sales and profits Tools: Advertise own product or industry as a whole Evaluation: **Natural experiments** (like the Super Bowl) enabled researchers to find out how effective advertising was in causing an increase in **market share**	Aim: Increase incentive to innovate Tools: Establish time-limited private monopolies Evaluation: Use **natural experiments** in history to find out whether **patents** and **copyright** encourage innovation, and how best to design these policies (breadth, length, etc.).

→ Find ways to identify **causes** rather than simply finding correlations.

F. Using Economic Models

How models are used in economics

- Models are an essential part of an economist's toolkit, ideally complemented by a narrative and supported by data:

		Bala	
		Rice	Cassava
Anil	Rice	Both produce rice: there is a glut of rice (low price) There is a shortage of cassava Anil not producing cassava, which he is better able to produce	No market glut High prices for both crops Both farmers producing the crop for which they are less suited
	Cassava	No market glut High prices for both crops Both farmers producing the crop for which they are better suited	Both produce cassava: there is a glut of cassava (low price) There is a shortage of rice Bala not producing rice, which he is better able to produce

Models come in various forms: diagrams, different types of games, mathematical equations, and so on.

"No economic model can be a perfect description of reality. But the very process of constructing, testing and revising models, forces economists and policymakers to tighten their views about how an economy works. This in turn promotes scientific debate over what drives economic behavior and what should (or should not) be done to deal with market failures." – Sam Ouliaris, IMF

A good model has five attributes:

- ✓ It is clear
- ✓ It identifies important relationships
- ✓ It predicts accurately
- ✓ It improves communication
- ✓ It is useful